Similarity-Cue Injection for Flexible Frame Selection in Video Question Answering

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GitHub: `https://github.com/manishhingar28/DLCV_COURSE_PROJECT`

I. Abstract (Maximum 100 Words)

1. Video-language models are costly because every input frame passes through a large model, so systems feed a few uniformly sampled frames and can miss the moment the question is about.
2. Buch et al. (CVPR 2025) put a light Flexible Frame Selector (FFS) in front of the large model: it scores frames against the question and keeps the useful ones through a differentiable top-k.
3. Learnable `[CTRL]` tokens inside that top-k let it also decide *how many* frames ($k-c$ to k) each question needs. It learns from the answer loss alone, with no frame labels.

II. Proposed Novelty & Technical Contribution

1. **Similarity-cue injection** (new module, 0.07M parameters): FFS must discover frame relevance from the answer loss alone, through a straight-through top-k. With frozen EVA02-CLIP-L/14 features this signal is weak (its learned top-4 is only +0.7 over uniform frames). We compute four *zero-shot relevance cues* per frame from the same frozen features (cosine to the question, best and second-best "question + option" similarity, and their gap), standardise them within each video, and feed them to the selector's frame tokens and, through a linear skip, to its selection logit.
2. It changes the selector input only: encoder, `[CTRL]` tokens, Gumbel top-k, losses and downstream model are unchanged. It needs no labels and no extra network.
3. Difference: the baseline learns question–frame similarity through attention; ours starts from a zero-shot ranker and learns a correction. Both share our FFS recipe (`[CTRL]` count hinge from fixed-k policies).

III. Theoretical / Mathematical Justification

1. $s_{mt} = \langle \hat{t}_m, \hat{f}_t \rangle$, $c_t = [\langle \hat{q}, \hat{f}_t \rangle, s_{(1)t}, s_{(2)t}, s_{(1)t} - s_{(2)t}]$, $\tilde{c}_t = (c_t - \mu_{\text{vid}})/\sigma_{\text{vid}}$, $l_t = g_\theta\big(h_t + \phi(\tilde{c}_t)\big) + w^\top \tilde{c}_t$, (hats: normalised features; $\hat{t}_m$: text of option m; $s_{(1)}, s_{(2)}$: two best options). For $w{=}0, \phi{=}0$ the model equals FFS, so the proposed model class contains the baseline.
2. **Residual learning:** without cues, relevance must be learned as a bilinear map $\langle W_Q \hat{q}, W_K \hat{f}_t \rangle$ in a 768-d space from noisy straight-through gradients. Ours supplies this statistic in closed form ($g_\theta{=}0$ ranks by $w^\top \tilde{c}_t$), so g_θ only learns a correction, which lowers sample complexity.
3. **Margin = evidence:** with $p(m|f_t) \propto e^{s_{mt}/T}$, a large gap $s_{(1)t} - s_{(2)t}$ means a low posterior entropy $H(a \,|\, q, f_t)$, so the cue marks frames that *discriminate* between answers, without the answer label.
4. **Standardisation:** top-k depends only on the order of l_t, which no per-video affine change alters; CLIP cosines shift strongly across videos (mean frame–frame cosine 0.89), so $\tilde{c}_t$ removes this nuisance.
5. **Ablation hypothesis (RAS):** a redundancy-aware greedy top-k, $i_t = \arg\max_i l_i - \beta \sum_{j \in A_{t-1}} S_{ij}$ (submodular), on the idea that near-duplicate frames waste the budget.

IV. Comparative Results & Empirical Evaluation

1. NExT-QA multiple choice, val (4,996 Qs), 32-frame pool, frozen EVA02-CLIP-L/14 features; a small frozen QA model (2.0M params) replaces the paper's 8B VLM, so absolute accuracy is not comparable to the paper's.
2. Same downstream model and recipe for all; tuning on held-out *training* videos only; 3 seeds, mean ± std; Cost = frames given to the downstream model / 32.
3. Takeaway: +1.05 accuracy at 4 frames and +0.85 at 1 frame for +0.07M parameters; 4 selected frames beat 32 uniform frames at 1/8 of the cost.

Table 1: Quantitative comparison of the baseline and proposed method on NExT-QA (val, 3 seeds). Best results are highlighted in bold.

Method	Frames	Acc (% ↑)	Causal	Temp.	Descr.	Cost (↓)	Params (M)
Uniform@32 / Uniform@4	32 / 4	48.40 / 47.94	46.9 / 46.8	46.3 / 45.1	57.9 / 57.7	1.000 / 0.125	–
Baseline: Top-4 (FFS-style input)	4	48.66 ± 0.19	47.8	45.7	57.7	0.125	2.05
Ours: Top-4 + cues	4	**49.71 ± 0.10**	**48.3**	**47.4**	**59.1**	0.125	2.12
Baseline: FFS [paper], adaptive	1.0	46.35 ± 0.19	44.9	44.0	56.2	**0.031**	2.05
Ours: FFS + cues, adaptive	1.0	47.20 ± 0.10	45.4	45.0	57.6	**0.031**	2.12
Ablation: Top-4 + cues + RAS	4	49.71 ± 0.10	48.3	47.3	59.3	0.125	2.12
Δ (Ours – Baseline), top-4 / FFS	–	+1.05 / +0.85	+0.5 / +0.6	+1.7 / +1.0	+1.4 / +1.4	0	+0.07

Result Interpretation: Cues improve both selectors in every question type (most: Temporal), by about 5× the seed std. With cues, 4 selected frames *beat* all 32 uniform frames (+1.3) at 1/8 of the cost; the FFS-style input only matches them (+0.3). FFS+cues with one frame ties Uniform@2 (47.14) at half the cost. Adaptive selectors settled on one frame, as this downstream model gains only 3.0 points from 1→32 uniform frames. RAS left the picks unchanged (score gaps dwarf the penalty); learned picks are *more* similar than uniform ones (0.92 vs 0.89) yet more accurate, so evidence clusters around the key moment and diversity is not the bottleneck. Caveats: no per-question significance test; a small downstream model, not an 8B VLM; the selector reads features of all 32 frames (cost counts downstream frames, as in the paper).